

## **LECTURE 10: Case Study**

Module: CT127-3-2 Programming for Data Analysis

Duration: 4 hours

Lecturer: Dr. Kulothunkan Palasundram (Dr. Kulo)

### **Learning Outcomes**

By the end of this lecture, students should be able to:

- Perform EDA and Advance Analysis
- Write clear, testable, and measurable hypotheses based on their analysis
- Perform hypothesis testing
- Interpret the result and make conclusions from hypothesis testing

## Case Study: Dataset: PimaIndiansDiabetes (built in dataset)

```
# Load required libraries
library(MASS) #dataset
library(ggplot2)
library("dplyr")

# Load the Pima Indians Diabetes dataset
data(Pima.tr)
data(Pima.te)
pima <- rbind(Pima.tr, Pima.te)
head(pima)

# Rename the 'type' column to 'diabetes' for clarity
names(pima)[names(pima) == 'type'] <- 'diabetes'
head(pima)
```

```
> head(pima)
  npreg glu bp skin  bmi  ped age diabetes
1     5  86 68  28 30.2 0.364  24       No
2     7 195 70  33 25.1 0.163  55      Yes
3     5  77 82  41 35.8 0.156  35       No
4     0 165 76  43 47.9 0.259  26       No
5     0 107 60  25 26.4 0.133  23       No
6     5  97 76  27 35.6 0.378  52      Yes
```

**Objective 1:** I intend to study about the effect of Glucose level to determine its effect on individuals being diagnosed with diabetes

## Analysis Questions:

AQ1: What is difference in mean and median of glucose levels for individuals with diabetes and without diabetes?

AQ2: Does individuals with high glucose level (higher than mean or median) have significantly higher proportion of the risk of getting diabetes?

## Analysis for AQ1:

AQ1: What is difference in mean and median of glucose levels for individuals with diabetes and without diabetes?

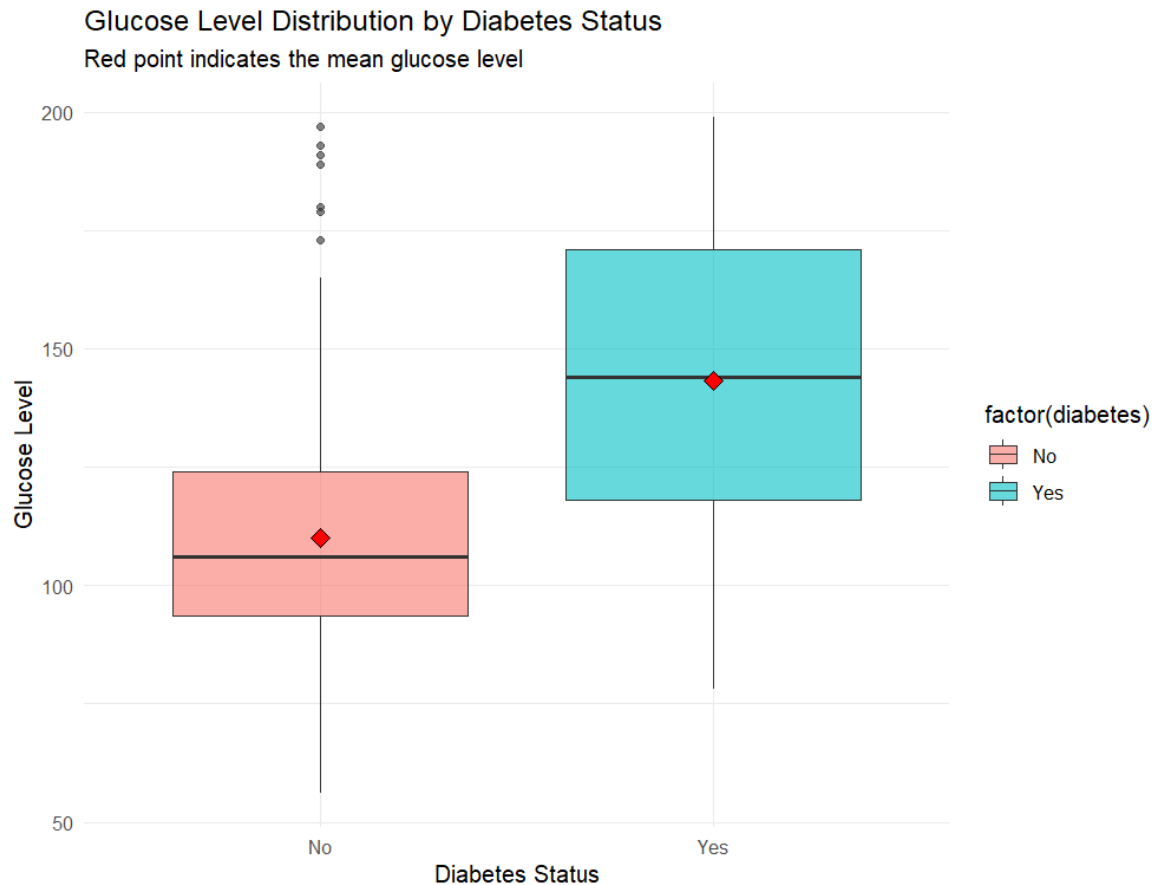
### Summary

```
> summary_data <- pima %>%
+   group_by(diabetes) %>%
+   summarise(
+     mean_glucose = mean(glu, na.rm = TRUE),
+     median_glucose = median(glu, na.rm = TRUE),
+     .groups = "drop"
+   )
> print(summary_data)
# A tibble: 2 × 3
  diabetes mean_glucose median_glucose
  <fct>      <dbl>         <int>
1 No          110.           106
2 Yes         143.           144
> |
```

Interpretation: Summary table above shows the average and median levels of individual with and without

diabetes. It can be seen that individuals with diabetes has much higher mean and median values compared to individuals without diabetes.

## Visualization



**Interpretation:** Box plot above shows the distribution of glucose levels for individuals with diabetes and without diabetes. Plot shows that individual with diabetes has much higher average and median of glucose levels compared to individuals without diabetes.

## Hypothesis Formulation:

Based on the 2 analyses above, we can formulate our hypothesis as follows:

**H<sub>0</sub>** = There is no significant difference between the **mean** glucose level of individuals with diabetes and without diabetes.

**H<sub>1</sub>** = The **mean** glucose level of individuals with diabetes is significantly higher than without diabetes.

Or

**H<sub>0</sub>** = There is no significant difference between the **median** glucose level of individuals with diabetes and without diabetes.

**H<sub>1</sub>** = The **median** glucose level of individuals with diabetes is significantly higher than without diabetes.

## Hypothesis Testing

Check for normality for glucose

```
> shapiro_test <- shapiro.test(pima$glu)
> shapiro_test
```

```
Shapiro-Wilk normality test
```

```
data: pima$glu
W = 0.95784, p-value = 3.345e-11
```

p-value < 0.05 means data is not normal.

**Hypothesis Testing Technique:** Since data is not normally distributed and involves one numeric variable and one categorical variable, we choose Wilcoxon Rank test

```
> # Ensure diabetes is a factor
> pima$diabetes <- factor(pima$diabetes, levels = c("No", "Yes"))
>
> # Split glucose by diabetes status
> glucose_yes <- pima$glu[pima$diabetes == "Yes"]
> glucose_no <- pima$glu[pima$diabetes == "No"]
>
> wilcox.test(glucose_yes, glucose_no,
+             alternative = "two.sided",
+             exact = FALSE)

      Wilcoxon rank sum test with continuity correction

data:  glucose_yes and glucose_no
W = 49890, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

p-value < 0.05 means we have sufficient evidence to reject the null hypothesis.

## Analysis for AQ2:

AQ2: Does individuals with high glucose level (higher than mean or median) have significantly higher proportion of the risk of getting diabetes?

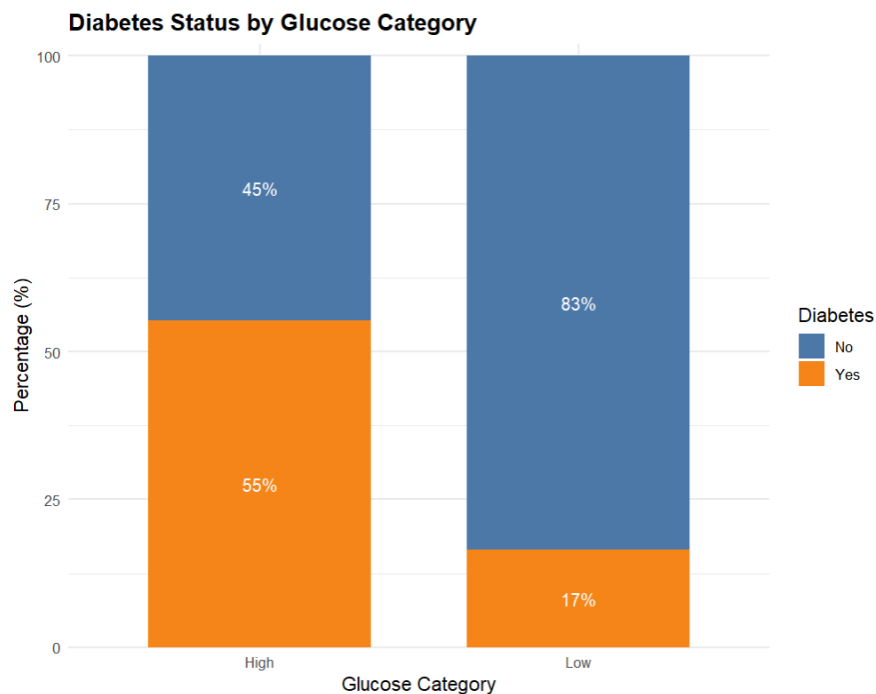
Note: To find answer for this question, we need to first create a new column to store the binary value for glucose level which are low and high. You can choose

either the mean or median value. In this case, I am using mean.

```
> # Create categorical variable for glucose level
> pima$glucose_cat <- ifelse(pima$glu >= mean(pima$glu, na.rm=TRUE), "High", "Low")
> head(pima)
  npreg glu bp skin  bmi  ped age diabetes glucose_cat
1     5  86 68  28 30.2 0.364 24      No      Low
2     7 195 70  33 25.1 0.163 55     Yes     High
3     5  77 82  41 35.8 0.156 35      No      Low
4     0 165 76  43 47.9 0.259 26      No     High
5     0 107 60  25 26.4 0.133 23      No      Low
6     5  97 76  27 35.6 0.378 52     Yes      Low
>
> # Summary of glucose level vs diabetes status
> table_summary <- table(pima$glucose_cat, pima$diabetes)
> table_summary

      No Yes
High 103 127
Low  252  50
>
> # Row-wise percentage
> prop_table <- prop.table(table_summary, margin = 1) * 100
> round(prop_table, 0)

      No Yes
High  45  55
Low   83  17
```



Interpretation (summary table and chart): The summary table and column chart shows that higher proportion (55%) of individuals with High glucose level is diagnosed with diabetes compared to only 17% of individuals with Low glucose level.

### **Hypothesis Formulation:**

$H_0$  = There is no difference on the likelihood to be diabetic between individuals with high and low glucose level.

$H_1$  = Individuals with high glucose level has **higher likelihood** to be diabetic than individual with low glucose level.

### **Hypothesis Testing**

Technique: Since both are categorical variable, we choose Chi-Square Test of Independence



```
> tab <- table(pima$glucose_cat, pima$diabetes)
> tab

      No Yes
High 103 127
Low  252  50

>
> chisq_test <- chisq.test(tab)
> chisq_test

      Pearson's Chi-squared test with Yates' continuity correction

data:  tab
X-squared = 86.168, df = 1, p-value < 2.2e-16
```

p-value < 0.05 means we have sufficient evidence to reject the null hypothesis.

**Objective 2:** I want to study the association of BMI categories (obese and non-obese) and diabetes status. BMI values will be categorized to obese status (obese and non-obese) based on the definitions in WHO website: [https://www.who.int/health-topics/obesity#tab=tab\\_1](https://www.who.int/health-topics/obesity#tab=tab_1)

Overweight and obesity are defined as abnormal or excessive fat accumulation that presents a risk to health. A body mass index (BMI) over 25 is considered overweight, and over 30 is obese. In 2019, an estimated 5 million noncommunicable disease (NCD) deaths were caused by higher-than-optimal BMI.

## Analysis Questions:

AQ1: Is there any kind of association between obese status and diabetes status?

AQ2: If yes, is it positively related?

AQ3: How strong is the relationship between obese and diabetes status?

## Analysis:

### Prepare data for analysis

```
> # Create binary BMI variable
> pima$obese <- ifelse(pima$bmi >= 30, "Obese", "Non-obese")
> pima$obese <- factor(pima$obese, levels = c("Non-obese", "Obese"))
> head(pima)
```

|   | npreg | glu | bp | skin | bmi  | ped   | age | diabetes | glucose_cat | obese     |
|---|-------|-----|----|------|------|-------|-----|----------|-------------|-----------|
| 1 | 5     | 86  | 68 | 28   | 30.2 | 0.364 | 24  | No       | Low         | Obese     |
| 2 | 7     | 195 | 70 | 33   | 25.1 | 0.163 | 55  | Yes      | High        | Non-obese |
| 3 | 5     | 77  | 82 | 41   | 35.8 | 0.156 | 35  | No       | Low         | Obese     |
| 4 | 0     | 165 | 76 | 43   | 47.9 | 0.259 | 26  | No       | High        | Obese     |
| 5 | 0     | 107 | 60 | 25   | 26.4 | 0.133 | 23  | No       | Low         | Non-obese |
| 6 | 5     | 97  | 76 | 27   | 35.6 | 0.378 | 52  | Yes      | Low         | Obese     |

Since both variables (IV and DV) are binary variables, we can compute the Phi Coefficient to identify the correlation between them.

```
> library(DescTools)
>
> tab <- table(pima$diabetes, pima$obese)
> DescTools::Phi(tab) # returns phi
[1] 0.275102
```

## Interpretation

Phi coefficient value of 0.28 shows a **positive association** between diabetes (Yes/No) and obese (Obese/Non-Obese). In other words, individuals classified as **Obese** are more likely to have **diabetes = Yes** compared to Non-obese individuals. However, it is a **weak** association.

## Hypothesis Formulation:

**H<sub>0</sub>** = There is **no association** between obese status and diabetes status.

**H<sub>1</sub>** = There **is an association** between obese status and diabetes status.

## Hypothesis testing

```
> # Create binary BMI variable
> pima$obese <- ifelse(pima$bmi >= 30, "Obese", "Non-obese")
> pima$obese <- factor(pima$obese, levels = c("Non-obese", "Obese"))
> # Contingency table
> tab <- table(pima$obese, pima$diabetes)
> tab
```

|           | No  | Yes |
|-----------|-----|-----|
| Non-obese | 157 | 29  |
| Obese     | 198 | 148 |

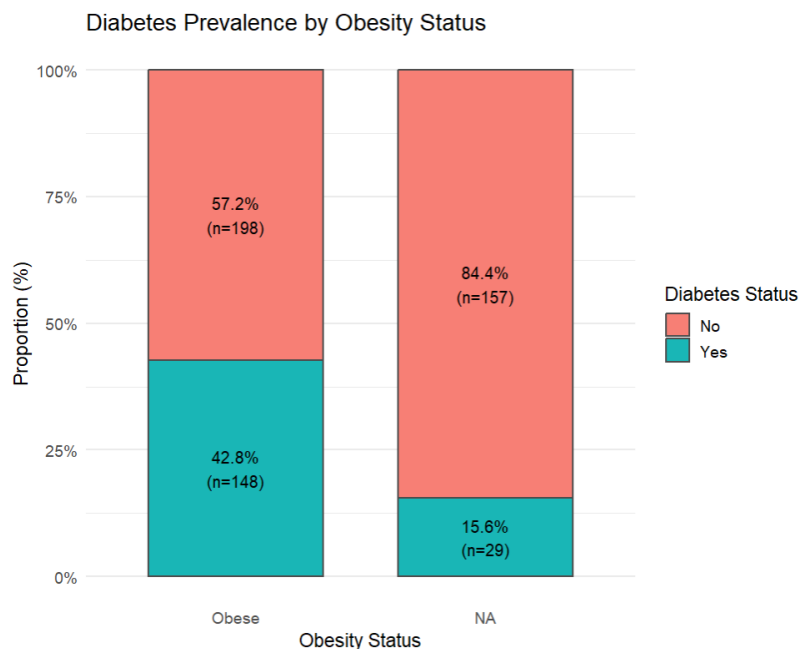
```
>
> chi_test <- chisq.test(tab)
> chi_test
```

Pearson's Chi-squared test with Yates' continuity correction

data: tab  
X-squared = 39.047, df = 1, p-value = 4.137e-10

- Since the p-value is far less than 0.05, we reject the null hypothesis ( $H_0$ ).
- Conclusion: There is a statistically significant association between obese status and diabetes status.

### Further Analysis by Visualization



**$H_0$**  = Obese individuals do not have a significantly higher risk of diabetes compared to Non-obese individuals.

**$H_1$**  = Obese individuals have a significantly higher risk of being diagnosed with diabetes compared to Non-obese individuals.

## Hypothesis Testing:

```

/
> # Contingency table
> tab <- table(pima$obese, pima$diabetes)
> tab

              No Yes
Non-obese 157  29
Obese     198 148
> fisher.test(tab, alternative = "greater")

      Fisher's Exact Test for Count Data

data:  tab
p-value = 4.78e-11
alternative hypothesis: true odds ratio is greater than 1
95 percent confidence interval:
 2.719551      Inf
sample estimates:
odds ratio
 4.036371

```

Interpretation: The odds of diabetes among obese individuals are about 4 times higher than among non-obese individuals

## Complex Hypothesis Formulation:

A complex hypothesis can be also formed from several simple hypotheses.

For example, let's consider our previous 2 hypotheses as listed below:

First hypothesis – using glucose level

**H<sub>0</sub>** = There is no difference on the likelihood to be diabetic between individuals with high and low glucose level.

**H<sub>1</sub>** = Individuals with high glucose level has higher likelihood to be diabetic than individual with low glucose level.

### Second hypothesis – using obesity level

**H<sub>0</sub>** = Obese individuals do not have a significantly higher risk of diabetes compared to Non-obese individuals.

**H<sub>1</sub>** = Obese individuals have a significantly higher risk of being diagnosed with diabetes compared to Non-obese individuals.

### Combined hypothesis

**H<sub>0</sub>** = Obese individuals with high glucose levels does not show any significant difference in risk of getting diabetes compared to Non-obese individuals or individuals with low glucose level.

**H<sub>1</sub>** = Obese individuals with high glucose levels have a significantly higher risk of getting of diabetes compared to Non-obese individuals or individuals with low glucose level.

## Hypothesis Testing:

To test the complex hypothesis which states interaction between 2 independent variables, we can use GLM (generalised linear model):

```
model <- glm(diabetes ~ obese * glucose_cat,
             data = pima,
             family = binomial)
```

```
summary(model)
```

```
Call:
glm(formula = diabetes ~ obese * glucose_cat, family = binomial,
    data = pima)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      -2.2858     0.3029  -7.545 4.52e-14 ***
obeseObese         1.0255     0.3543   2.894 0.003801 **
glucose_catHigh    1.4554     0.4198   3.467 0.000527 ***
obeseObese:glucose_catHigh 0.3464     0.4845   0.715 0.474598
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 676.79  on 531  degrees of freedom
Residual deviance: 559.36  on 528  degrees of freedom
AIC: 567.36

Number of Fisher Scoring iterations: 4
```

## Interprteation:

- i) Being obese increases log-odds of diabetes by 1.026 or  $e^{1.026} \approx 2.79$ , which means that Obese individuals with normal glucose have  $\sim 2.8$  times higher odds of diabetes than non-obese

individuals.

- ii) High glucose increases log-odds by 1.455 or  $e^{1.455} \approx 4.29$  which means that High glucose increases diabetes odds by  $\sim 4.3$  times, even for non-obese individuals.
- iii) But the interaction term `obeseObese:glucose_catHigh` has p-value = 0.475 which means that it is not significant, or in other words the effects of obesity and high glucose on diabetes are additive, not multiplicative and high glucose increases diabetes risk similarly for obese and non-obese individuals